

An Overall Distribution Particle Swarm Optimization MPPT Algorithm for Photovoltaic System Under Partial Shading

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Abstract—Solar photovoltaic (PV) systems under partial shading conditions (PSCs) have a nonmonotonic P - V characteristic with multiple local maximum power points, which makes the existing maximum power point tracking (MPPT) algorithms unsatisfactory performance for global MPPT, if not invalid. This paper proposes a novel overall distribution (OD) MPPT algorithm to rapidly search the area near the global maximum power points, which is further integrated with the particle swarm optimization (PSO) MPPT algorithm to improve the accuracy of MPPT. Through simulations and experimentations, the higher effectiveness and accuracy of the proposed OD-PSO MPPT algorithm in solar PV systems is demonstrated in comparison to two existing artificial intelligence MPPT algorithms.

Index Terms—Maximum power point tracking (MPPT), particle swarm optimization (PSO), partial shading conditions (PSCs), solar photovoltaic (PV) systems.

I. INTRODUCTION

IN RECENT years, the rapid consumption of fossil fuels has resulted in severe environmental pollution, making renewable energy sources a more favorable option in generating electricity due to their inexhaustibility and environmental

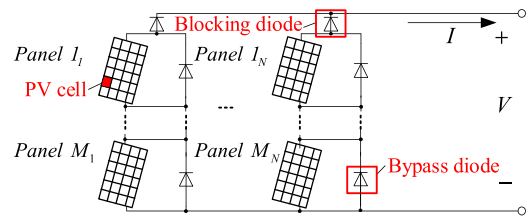


Fig. 1. Structure of the PV array.

friendliness. Solar energy is considered one of the most important renewable energy sources and has been widely used in photovoltaic (PV) power generation [1]–[5]. However, the low efficiency and high cost of PV systems impede the development of PV power generation. Since the output characteristics of PV panels are influenced by external factors, such as solar irradiance and temperature, the maximum power point (MPP) of a PV panel will vary along with these external factors. Therefore, an effective maximum power point tracking (MPPT) technique is of great importance for efficiency improvement of PV power generation systems.

In a PV system, each PV array consists of many PV panels connected in series and parallel to exert high voltage and high current, for the purpose of increasing output power of PV arrays, as shown in Fig. 1. There are two types of diodes in a solar power system, which are bypass diodes and blocking diodes for different purposes. Blocking diodes prevent backflow of electricity. Bypass diodes are used to inhibit hotspot heating effect and reduce the power loss due to shading. This situation is defined as PSC where each panel may receive and experience different solar irradiances and temperatures at the same time [5]–[9]. When a PV array is under PSCs, its P - V (power-voltage) characteristic curve will have multiple peaks. Among all the peaks, there is only one global MPP (GMPP), and other peaks are called local MPPs (LMPPs). In the literature, many MPPT algorithms [11]–[29] have been proposed to track the GMPP of PV systems under PSCs.

It is well-understood that the traditional MPPT algorithms, such as perturbation and observation (P&O) algorithm, incremental conductance (IC) algorithm, and constant voltage (CV) algorithm, cannot well track the GMPP under PSC [2]. To solve the GMPP problem, many other algorithms have been proposed. The algorithm presented in [10] is an improvement of

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the IC algorithm, which can find the GMPP by identifying all LMPPs and GMPP, but it requires a wide range of information about the PV arrays, such as open-circuit voltage and short-circuit current of each PV panel. The GMPPT algorithm proposed in [5] is to scan the P - V curve by increasing the output voltage of PV array step by step, which is helpful to improve the transient response of the PV system and increase the tracking speed. However a high-precision, fast-speed digital processor is required for the implementation of this algorithm, which will increase the cost of the overall PV system. The proposed MPPT algorithm in [11] is to first estimate the possible GMPP according to the open-circuit voltage of the PV array and then use the P&O-based MPPT algorithm to track the GMPP. The MPPT algorithm in [11] has good dynamic tracking performances; however, due to its algorithmic nature, this algorithm could be invalid for GMPP under complex PSCs. In [12]–[16], the regions of all MPPs are first calculated to decrease the tracking time, which however requires excessive information about the structure of PV array. These values are difficult to obtain or predict in real-world engineering practices.

Furthermore, MPPT can be considered an optimization problem which can be solved by using AI-based MPPT algorithms [17]–[26]. Popular AI-based MPPT algorithms used for MPPT include particle swarm optimization algorithm, firefly algorithm and ant colony optimization algorithm. Compared with the traditional MPPT methods, the main advantage of AI-based MPPT algorithms is that these algorithms can find the GMPP under PSCs without a global scan of P - V curve, which requires less sampling points in the process and thus entails lower computational costs. The PSO algorithm has been widely used for MPPT due to its simplicity in mathematical structure and implementation [22]–[24]. In [23], the problems of nonconvergence and failure to capture the GMPP in the PSO algorithm have been pointed out, and a GMPPT method was proposed based on power closed-loop, which however could increase the complexity of the control design. In [18], an improved firefly MPPT algorithm was proposed to increase the MPPT speed by simplifying the movement rules of fireflies in the algorithm. In recent years, the simulated annealing algorithm and ant colony algorithm have also been proposed for the MPPT [25], [26] in PV systems. In [36] and [37], improved Java MPPT algorithms were proposed without algorithm-specific parameters.

According to the experimental results in [25] and [26], suitable initial values are very important for improving the tracking speed of the AI-based MPPT algorithms. The determination of a small region containing the GMPP, with which the initial values of AI-based MPPT algorithms can be assigned, is thus important to increase the tracking speed [24], [28], [29]. Several strategies in finding this region are documented in [24], [28], and [29], where however the detailed information of each PV panel, such as the open-circuit voltage and short-circuit current, is needed. This may reduce the feasibility of implementing these algorithms into hardware, due to the necessity of preimplementation hardware testing.

Having well understood the drawbacks of these aforementioned existing MPPT algorithms, in this paper, we propose a novel overall distribution OD-PSO MPPT method, which does

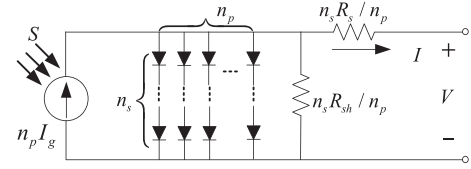


Fig. 2. Equivalent circuit model of a PV panel.

not require any hardware information of PV systems and is able to search and find the GMPP accurately and rapidly. Particularly, the OD MPPT algorithm is employed to locate the vicinity of the GMPP region, which facilitates the setting of initial values that will be imported to the PSO MPPT algorithm. After obtaining the initial values, the PSO MPPT algorithm will only need to find the GMPP within a very small search region, enabling it to find the exact GMPP in a rapid manner. Therefore, the integration of the OD MPPT algorithm and PSO MPPT algorithm is capable of quickly and accurately tracking and identifying the GMPP of PV panels in diverse PSCs.

The remainder of this paper is organized as follows. The output characteristics of the PV arrays and mathematical modeling under PSC are introduced in Section II. The OD-PSO MPPT algorithm is proposed in Section III, followed by its operating principle. In Section IV, MPPT simulation results using the proposed OD-PSO MPPT and other AI-based algorithms are presented, compared, and analyzed. Experimental results are provided in Section V. Finally, conclusions are drawn in Section VI.

II. CHARACTERISTICS OF PV ARRAY UNDER PSC

Suppose an $M \times N$ PV array is composed of M PV panels connected in parallel and N PV panels connected in series to obtain high current and high voltage in an electric circuit, as shown in Fig. 1. A PV panel is composed of a number of PV cells, which can be represented as an equivalent circuit, as shown in Fig. 2.

According to Fig. 2, the I - V characteristics of a PV panel can be expressed by (1) [28], [29], where A is the ideal diode factor, k is Boltzmann's constant, T is the temperature in Kelvin, q is the electron charge, R_s and R_{sh} are the equivalent series and shunt resistances, and I_{sc} is the saturation current of the diode. V and I are the output voltage and current of PV array:

$$I = n_p I_g - n_p I_{sc} \left\{ \exp \left[\frac{q \left(V + I R_s \cdot \frac{n_s}{n_p} \right)}{n_s A k T} \right] - 1 \right\} - \frac{n_p V / n_s + I R_s}{R_{sh}}. \quad (1)$$

Under PSC, the shaded panels cannot produce power and become a load, which may create hotspot heating effect and even cause damage to the entire PV panel [28]. To solve the above problem, a PV panel is normally connected in parallel to a bypass diode, as shown in Fig. 1. Therefore, the shaded panels can be bypassed and then be protected by the antiparalleled diode under PSC, which results in the multiple peaks in P - V characteristic

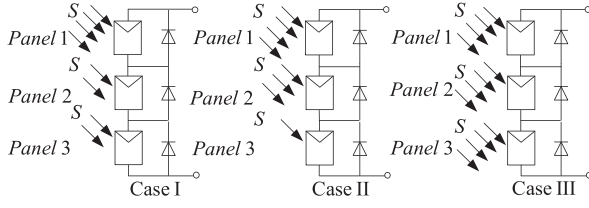
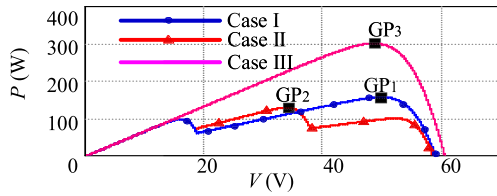


Fig. 3. PV array composed of three panels in series.

TABLE I
IRRADIANCES IN THIS STUDY

Situation	Irradiance of PV panels (W/m ²)		
	Panel 1	Panel 2	Panel 3
Case I	1000	500	500
Case II	1000	600	300
Case III	1000	1000	1000

Fig. 4. P - V characteristic curves of the PV array under PSC.

curve of PV array [8]. Furthermore, the existing results have shown that the multiple peaks in P - V characteristic are related to the PV panels connected in series in the PV array under PSCs [8]. Therefore, in this paper, a PV array composed of three panels in series is chosen as the testbed to validate the proposed OD-PSO MPPT algorithm. Three cases with different solar irradiance of the PV array are illustrated in Fig. 3. The irradiances are shown in Table I. Case III is used as a control group where the irradiance is identical for all solar panels. The P - V characteristics under these three cases, shown in Fig. 3, are simulated in PSIM, and the simulation results are shown in Fig. 4.

It is obvious that in Case I, PV array receives two different radiation intensities; the irradiance of the shaded PV panels is only 400 W/m². There is a voltage difference between panel 1 and panel 2 (or panel 3), which will turn on the bypass diodes of panel 2 and panel 3 [8]. Thus, there will be two MPPs on the P - V characteristic of the PV array, as shown in Fig. 4. In this case, the GMPP is located at GP₁. In Case II, three PV panels of the PV array receive three different irradiances, which leads to three MPPs on the P - V characteristic curve and the GMPP is now located at GP₂, as shown in Fig. 4. In case III, PV array works without any shading, so in Fig. 4 there is only one power peak on the P - V characteristic curve, namely GP₃.

It can be seen that both the number of the MPPs and the peak values of the P - V characteristic differ from case to case. Therefore, it is very important to identify the GMPP for PV systems under PSCs.

III. PRINCIPLE OF THE OD-PSO MPPT ALGORITHM

Although the AI-based MPPT algorithms can be used to accurately capture the GMPP. They are sensitive to initial condition.

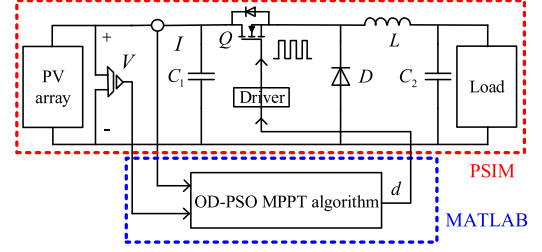


Fig. 5. Small PV power generation system.

Optimizing the initial values using a variety of measures [25], [26] can improve the performance of the AI-based MPPT algorithms. In the following, an OD MPPT algorithm will be proposed to shrink the search region of GMPP, which obtains the initial particles for the AI-based MPPT algorithms. Therefore, the two algorithms are able to complement each other and achieve a better MPPT performance.

A. OD MPPT Algorithm

The OD algorithm is an intelligent algorithm that was first proposed in 2012 [30]. The OD algorithm has been successfully applied to solve the short-term optimization dispatch problems for hydrothermal power systems [30], [31]. In this paper, the OD MPPT algorithm is employed to rapidly find the rough search region around GMPP. The mathematical expression of the OD MPPT algorithm is given in the following equation:

$$\begin{cases} R^{j+1} = \alpha R^j \\ d_i^{j+1} = \text{mppt_gbest}^j + R^{j+1} \text{Cauchy}_i^{j+1} \\ \text{Cauchy}_i^{j+1} = -\gamma \tan(\pi r_1) \end{cases}$$

$$i = 1, 2, \dots, N, \text{Cauchy} \in [-2, 2], \alpha \in (0, 1). \quad (2)$$

Term d_i^{j+1} is the position of the i th particle at the $(j+1)$ th iteration, R^j is the Cauchy radius in j th iteration, α is the shrinking coefficient, Cauchy_i^{j+1} is the Cauchy distribution random number of particle i at the $(j+1)$ th iteration, and γ is the scale parameter is a random number.

According to the OD algorithm and (2), particles d_i^j can be set in a certain area around the GMPP directly due to Cauchy distribution. Furthermore, this region can reduce adaptively and the OD MPPT algorithm will end until the region is smaller than ε ($\varepsilon > 0$).

To introduce the OD MPPT algorithm, a small PV power generation system with a PV array and a buck converter is taken as an example, which is shown in Fig. 5. In this experiment, we model the inverter as a resistive load, without using an actual inverter to simplify the experimental setup and loosen the assumptions. Modeling the inverter as a resistive load is to account for the power transmitted into the dc/ac inverter, which can sufficiently reflect the mechanism of the inverter for the purpose of this study [26], [27]. For this PV power generation system, the duty cycle d of the buck converter is chosen as the particle of the OD MPPT algorithm. Term R^j Cauchy radius represents the distribution region of the particles in the OD MPPT algorithm at the j th iteration. R^j is between 0 and 1. The positions of

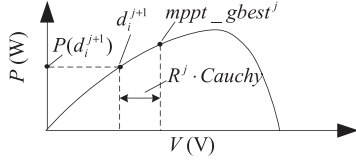


Fig. 6. P - V characteristic of the PV array.

particles are updated by R^j , Cauchy and mppt_gbest^j . The positions of particle d_i^j , Cauchy radius R^j , and the Cauchy distribution random number Cauchy of the OD MPPT algorithm can be defined by (2). Term mppt_gbest^j means the existing best position of particles in the OD MPPT algorithm within j iterations. The best position of particles indicates that the fitness value of the position is the maximum one among these particles. The fitness function of the PV power generation system can be expressed as follows:

$$P(d_i^{j+1}) = VI. \quad (3)$$

Terms V and I are the output voltage and current of PV array, respectively. In general, the P - V characteristic curve of PV array and the relationship among the parameters in the OD MPPT algorithm are depicted in Fig. 6.

The OD MPPT algorithm starts by initializing the duty cycles of the buck converter, which are noted as d_i^j ($i = 1, 2, \dots, N$). The corresponding power at each duty cycle can be calculated by (3). The maximum fitness value will be saved, and its corresponding duty cycle is set as d_{best}^j . According to the definition of mppt_gbest^j , mppt_gbest^j needs to satisfy the following relation:

$$P(\text{mppt_gbest}^j) \geq P(d_{\text{best}}^k) \quad (k = 1, 2, \dots, j). \quad (4)$$

This generation of duty cycles will be updated to track the GMPP, and the updated duty cycles d_i^{j+1} ($i = 1, 2, \dots, N$) can be calculated by (2). The new duty cycles are distributed around mppt_gbest^j within a region of radius R ($R \in [0, 1]$), which follows the Cauchy distribution. R is reduced by a shrinking coefficient α . The above process is repeated until the radius $R < \varepsilon$ ($\varepsilon > 0$) is satisfied.

In the OD MPPT algorithm, Cauchy distribution, a continuous probability distribution, is selected to update the particles [30]. The Cauchy distribution can be expressed as [30]

$$F(\text{Cauchy}) = \frac{1}{\pi} \left(\arctan(\text{Cauchy} - x_0) / \gamma + \frac{\pi}{2} \right). \quad (5)$$

Term x_0 is the position parameter used to define the distribution peak position, and γ is the scale parameter.

Cauchy distribution has its own probability density function, which is given as (6) [30]. In this paper, x_0 is chosen as 0:

$$f(\text{Cauchy}; x_0, \gamma) = \frac{1}{\pi \gamma \left[1 + \left(\frac{\text{Cauchy} - x_0}{\gamma} \right)^2 \right]}. \quad (6)$$

Because the position of particles d_i^j in the OD MPPT algorithm is a linear change of Cauchy , d_i^j has the same distribution

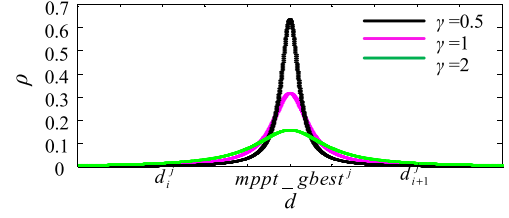


Fig. 7. Cauchy distribution function diagram under different γ .

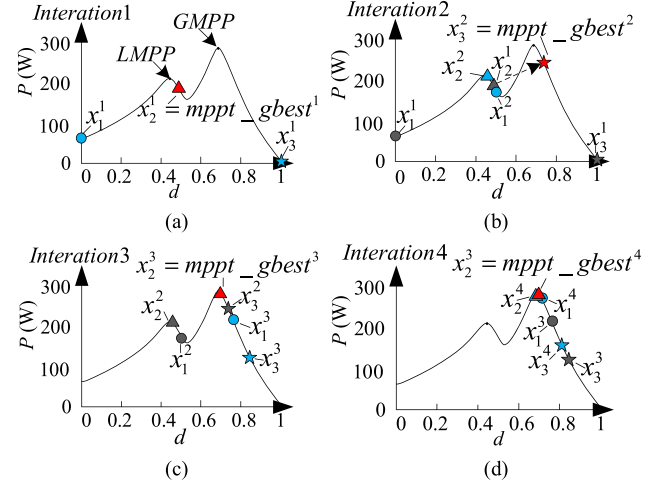


Fig. 8. Operating principle of the OD MPPT algorithm.

rule as Cauchy. The probability density ρ of Cauchy distribution is shown in Fig. 7.

It can be seen that the probability is more concentrated toward to the mppt_gbest^j when γ is smaller and the probability of particles in Cauchy distribution is more dispersed when γ is greater [30]. Therefore, the closer to the mppt_gbest^j , the higher the probability density of particles [30]. In the algorithm, the particles distributed near the mppt_gbest^j can be used to improve the accuracy of GMPP and particles far from the mppt_gbest^j can help the OD MPPT algorithm avoid tracking the LMPPs. Therefore, in theory, the OD MPPT algorithm has the ability to find the GMPP.

To better interpret the proposed OD MPPT algorithm, a non-monotonic P - d curve with two MPPs is given as an example, which is shown in Fig. 8, and the operating principle, i.e., the iteration steps, is given as follows.

First, let $N = 3$ for the OD MPPT algorithm. In general, GMPP will not occur at $d = 0$ or $d = 1$. Three initial particles are given as follows: $x_1 = 0.05$, $x_2 = 0.5$, $x_3 = 0.95$. In Fig. 8, x_1 , x_2 , x_3 are marked by a dot, triangle, and star, respectively. In simulation and experimentation, Cauchy_i^j ($i = 1, 2, 3$) is produced by (2). But now, in order to facilitate the description of the proposed algorithm, Cauchy_i^j ($i = 1, 2, 3$) are replaced by three fixed values, i.e., 0.03, -0.08, and 0.6, respectively. The initial value of Cauchy radius R^1 is usually set as half of interval of the initial particles. So, the initial value of R^1 is chosen to be $((0.95 - 0.05) / 2) = 0.45$.

Iteration 1: The OD MPPT algorithm sends these three initial particles to the buck converter and the corresponding output

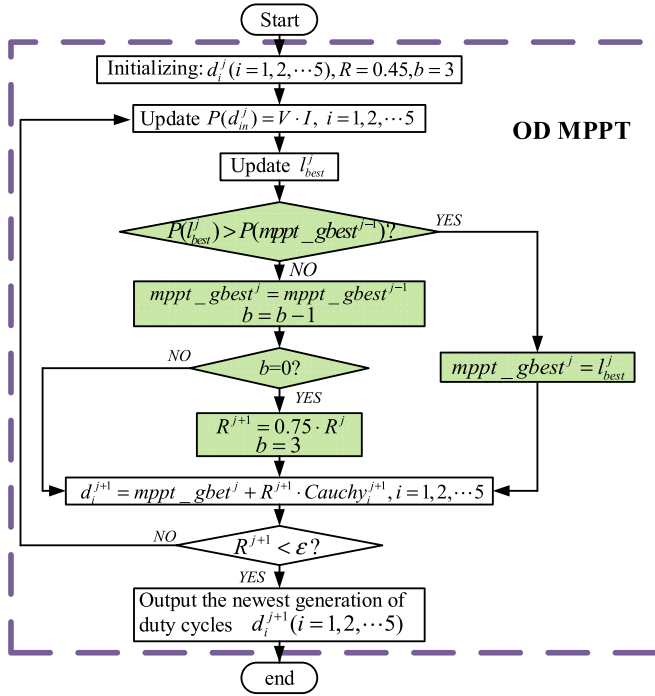


Fig. 9. Flow chart of the OD MPPT algorithm.

power of PV array will be calculated. As shown in Fig. 8(a), x_2^1 is the best particle of the present particles. Term x_2^1 is denoted as l_{best}^1 , and according to the definition in (4), l_{best}^1 is also $mppt_gbest^1$.

Iteration 2: In accordance with the obtained $mppt_gbest^1$, particles of the next generation are generated by (2). Particles are distributed as shown Fig. 8(b). It is obvious that x_3^2 is l_{best}^2 . Comparing l_{best}^2 with $mppt_gbest^1$, it can be found that l_{best}^2 is the best particles until now. Since $P(l_{best}^2) > P(mppt_gbest^1)$, $mppt_gbest^2$ will be replaced by l_{best}^2 and Cauchy radius R^j will decrease according to (2). If the shrinking coefficient α is set as 0.75 to meet the tracking performance, the Cauchy radius will be $R^2 = 0.45 \times 0.75 = 0.3375$ in this iteration.

Iteration 3: Repeat the process in iteration 2, and generate particles of the third generation, which are distributed as shown in Fig. 8(c). It is obvious that x_3^3 is the best particle at iteration 3, so l_{best}^3 is set as x_3^3 . Since $P(l_{best}^3) > P(mppt_gbest^2)$, $mppt_gbest^3$ is replaced by l_{best}^3 , the Cauchy radius will further shrink to $R^3 = 0.3375 \times 0.75$.

Iteration 4: The fourth-generation particles obtained by the same process are shown as Fig. 8(d). From Fig. 8(d), it can be seen that x_2^4 is the best particle in iteration 4. Therefore, l_{best}^4 is set as x_2^4 . Since $P(l_{best}^4) < P(mppt_gbest^3)$, $mppt_gbest^4$ is the same as $mppt_gbest^3$, the Cauchy radius now is $R^4 = 0.2531 \times 0.75 < \varepsilon$ ($\varepsilon = 0.2$).

According to the OD MPPT algorithm, the $mppt_gbest^4$ at the final iteration and other two particles will be considered as the optimal particles and $mppt_gbest^4$ will be regarded the G_{best}^1 . The flow chart of the OD MPPT algorithm is shown in Fig. 9.

In addition, in order to prevent the missing of GMPP, a stagnation coefficient b is added to the proposed OD MPPT algorithm, which is used to define the shrinking time of Cauchy

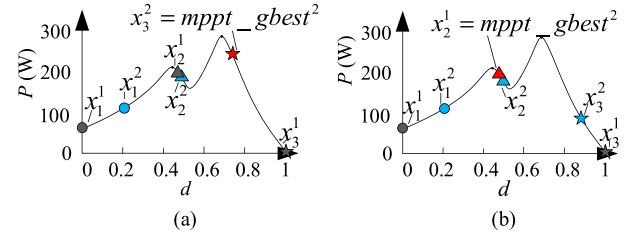


Fig. 10. Operating principle of the OD MPPT algorithm.

radius [30]. The increased values of stagnation coefficient are marked in different colors as shown in Fig. 9. Using Fig. 8(a) as an example, the duty cycles d_i^1 ($i = 1, 2, 3$) is updated by (2). The particles generated at the new generation d_i^2 ($i = 1, 2, 3$) may be distributed as Fig. 10(a) and may also be distributed as Fig. 10(b). In Fig. 10(a), the $mppt_gbest^2$ will change to x_3^2 , then the Cauchy radius remains unchanged and d_i^2 ($i = 1, 2, 3$) will be updated by (2). If the distribution of the three particles is shown as Fig. 10(b), the $mppt_gbest^2$ will not change to x_i^2 . Term $mppt_gbest^2$ remains the same as $mppt_gbest^1$. The stagnation coefficient b subtracts 1 and d_i^2 ($i = 1, 2, 3$) is updated accordingly. If the stagnation coefficient decreases to 0, the Cauchy radius is calculated as $R^2 = 0.75R^1$, and the stagnation coefficient b will be set to its original value. If the stagnation coefficient b does not decrease to 0, the Cauchy radius remains unchanged and d_i^2 ($i = 1, 2, 3$) is updated according to (2). Stagnation coefficient is used to find the best particles and ensure the location of GMPP can be found at a certain Cauchy radius. Normally, the stagnation coefficient is a positive integer, which can be set in the range from 3 to 5. A larger stagnation coefficient leads to a larger number of iterations, which will deteriorate the fluctuations in the output power of the PV system. Therefore, there is also a tradeoff between the tracking speed and tracking accuracy in the OD MPPT algorithm.

It should be noticed that, according to $R^{j+1} = \alpha R^j$ and $d_i^{j+1} = mppt_gbest^j + R^{j+1} \text{Cauchy}^j$, $R^{j+1} \text{Cauchy}^j$ will gradually reduce since $\alpha < 1$, and the generated particles will move closer to the obtained $mppt_gbest$ in each iteration. This means d_i^{j+1} will decrease over iterations and the GMPP speed will gradually slow down.

From the above analysis, it is obvious that the OD MPPT algorithm can find the small region containing the GMPP rapidly and do not need the information about the structure of PV array, such as open-circuit voltage and short-circuit currents of each PV panel. This feature can solve the problem mentioned in the Introduction for the improved AI-based MPPT algorithms. Due to the tracking effectiveness and structural simplicity of the PSO MPPT algorithm, it is selected to be integrated with the OD MPPT algorithm, which is branded as the OD-PSO MPPT algorithm in this paper.

B. OD-PSO MPPT Algorithm

The PSO algorithm was proposed in 1995 by Kennedy and Eberhart [32]. Later, the PSO algorithm has been widely used in optimal control designs, and the PSO MPPT algorithm was first proposed in [28]. The mathematical expression of PSO MPPT

algorithm is given in the following equation:

$$\begin{cases} v_i^{k+1} = \omega v_i^k + c_1 r_1 (P_{besti} - x_i^k) + c_2 r_2 (G_{best} - x_i^k) \\ x_i^{k+1} = x_i^k + v_i^{k+1} \end{cases} \quad (7)$$

Term k is the iteration number, ω is the inertia weight, r_1 and r_2 are random numbers within $[0, 1]$, and c_1 and c_2 are the cognitive coefficient and social coefficient, respectively. Term v_i^k represents the offset vector of the particle i at the k th iteration which is usually limited below a maximum v_{max} [28], and x_i^k represents the position vector of the particle i at the k th iteration. In each iteration, the position and the offset of the particles are updated by the personal best P_{besti} position of the particle and the global best G_{best} position of the particle swarm, respectively.

The position and offset of the particles are updated as follows [28]. From (7), it is obvious that the position of particles x_i^k is changed by the offset v_i^k to move closer to the GMPP. The PSO MPPT algorithm enables its particles to converge to the GMPP when $P_{besti} = G_{best}$. In order to avoid the particles of the PSO MPPT algorithm converging to an LMPP, the maximum offset v_{max} is always restricted to a small value [5], [33], and according to [5], v_{max} can be estimated by $v_{max} = 1/V_{oc}$, where V_{oc} is the open-circuit voltage of PV array. That is to say, the particle approaches the GMPP at v_{max} or a slower speed, which decreases the tracking speed of the PSO MPPT algorithm significantly. Therefore, although the PSO MPPT algorithm has high accuracy on tracking the GMPP [8], the tracking speed of the PSO MPPT algorithm is not satisfactory. With the initial particles obtained by the OD MPPT algorithm, the PSO MPPT algorithm can further be used to track the GMPP quickly and accurately.

In the OD-PSO MPPT algorithm, the OD MPPT algorithm is first used to rapidly obtain the particles, which are within a small region that contains the GMPP. The obtained particles will be used as the initial particles for the PSO MPPT algorithm, and finally the GMPP will be captured by the PSO MPPT algorithm. Since the GMPP of the PV array will change under different irradiances, the OD-PSO MPPT algorithm needs to restart to track the new GMPP. In order to verify the dynamic performances of the OD-PSO MPPT algorithm, the criterion for judging the extent of solar irradiance variation can be expressed as

$$\frac{|P_{n+1} - P_n|}{P_n} \geq \delta. \quad (8)$$

Term P_n denotes the n th output power of the PV array, δ is the threshold value, which is set to be 0.02 in this paper according to [5].

The flow chart of the OD-PSO MPPT algorithm is shown in Fig. 11. The flow chart of the OD MPPT algorithm is the same as shown in Fig. 9. Further, the term j_{max} is used to set the maximum iteration number of the PSO MPPT algorithm for eliminating the steady-state oscillation. Of course, too small j_{max} may also lead to accuracy problems. The choice of j_{max} requires a tradeoff between accuracy and suppression of steady-state oscillations.

From the above analysis, it can be seen that the OD-PSO MPPT algorithm has the following advantages.

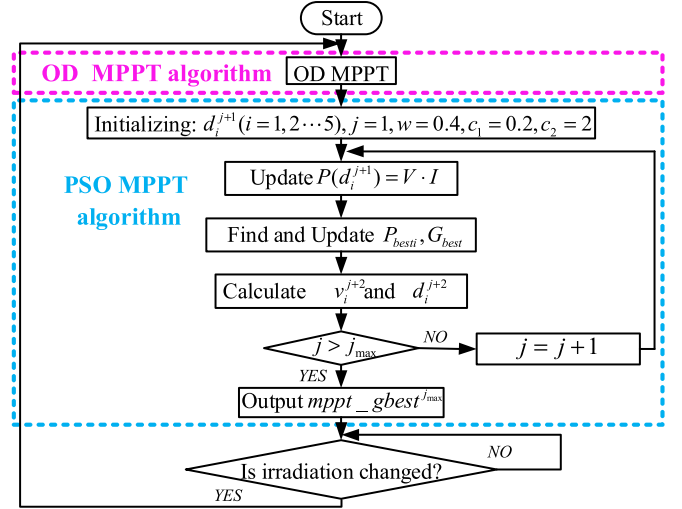


Fig. 11. Flow chart of the OD-PSO MPPT algorithm.

TABLE II
PARAMETERS FOR THE PV MODULES

Parameters	Values
Maximum power(P_{max})	101.3W
Open circuit voltage(V_{oc})	20.2V
Maximum power voltage(V_{mp})	16.7V
Short circuit current(I_{sc})	6.719A
Maximum power current(I_{mp})	6.074A
Current temperature coefficient	-0.379%/°C
Voltage temperature coefficient	0.0396%/°C

- 1) It can improve the tracking speed of MPPT compared to the AI-based MPPT algorithm.
- 2) It can rapidly find the small region that contains the GMPP without the information about the structure of PV array, which demonstrates a higher accuracy in finding GMPP.

IV. SIMULATION RESULTS

In order to validate the MPPT performance of the proposed OD-PSO MPPT algorithm, the simulation platform is established as depicted in Fig. 5 using PSIM and MATLAB. The duty cycle d of the buck converter is taken as the particle in the proposed OD-PSO MPPT algorithm. The circuit parameters of buck converter are as follows: $C_1 = 50 \mu\text{F}$, $C_2 = 22 \mu\text{F}$, $L = 600 \mu\text{H}$, $R = 1.7 \Omega$, and the switching frequency is 20 kHz. The parameters of the PV module are shown in Table II. In the simulation, ε is set as 0.035, and γ is set as 0.1 for the OD MPPT algorithm, by referring to [30]. The initial number of particles is 5, and the initial particles of OD-PSO MPPT algorithm are uniformly distributed within $[0, 1]$, which are chosen as $x_1 = 0.1$, $x_2 = 0.3$, $x_3 = 0.5$, $x_4 = 0.7$, $x_5 = 0.9$, and j_{max} is 10 for the PSO MPPT algorithm.

The parameters of the PSO MPPT algorithm are $\omega = 0.4$, $c_1 = 0.2$, $c_2 = 2$, and $v_{max} = 0.01$, to ensure that it converges to either LMPPs or GMPP [32]. The selection of cognitive and social parameters is adopted and modified from [34]. According

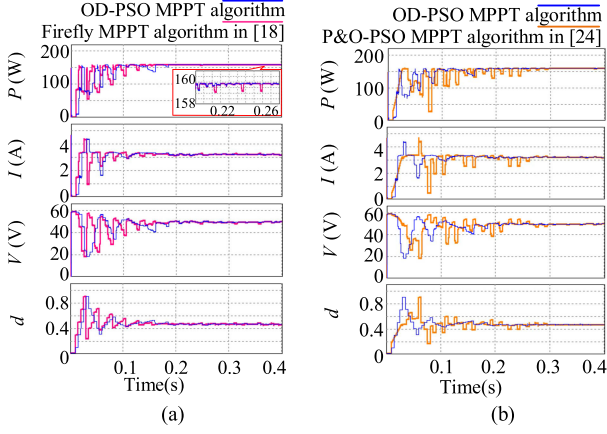


Fig. 12. Output waveforms of the PV system in Case I. (a) OD-PSO MPPT algorithm and the firefly MPPT algorithm in [18]. (b) OD-PSO MPPT algorithm and P&O-PSO MPPT algorithm in [24].

to [5], the sampling time of V and I needs to be longer than the transient response time of the PV system. The transient response time means the time duration that the converter stabilizes after a new duty cycle is implemented, and the transient response time is determined by the main circuit parameters. In this paper, the transient response time is within 1 ms, and therefore the sampling time is set to be 5 ms.

In order to compare the tracking performance of the proposed OD-PSO MPPT algorithm with other AI-based MPPT algorithms, the firefly MPPT algorithm in [18], denoted as the firefly MPPT algorithm in this paper, and the MPPT algorithm based on P&O combined with PSO in [24], denoted as the P&O-PSO MPPT algorithm in this paper, are implemented for comparison in this paper. The simulation results with the proposed OD-PSO MPPT algorithm, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm under different PSCs are carried out as follows.

Case I: The irradiance of PV panels is given in Table I. The P - V characteristic curve is shown in Fig. 4. From Fig. 4, it can be seen that the output voltage of the PV array for Case I at GP_1 is 49.86 V and the power of the PV array is 159.94 W.

The simulation results for Case I are shown in Fig. 12. In Fig. 12(a), the output waveforms of PV system using the proposed OD-PSO MPPT algorithm and the firefly MPPT algorithm are shown. With the proposed OD-PSO MPPT algorithm, it takes 0.210 s, eight iterations to successfully track the GMPP. Particularly, five iterations are used for the OD MPPT algorithm and three iterations for the PSO MPPT algorithm. With the firefly MPPT algorithm, it takes 0.206 s, eight iterations. In Fig. 12(b), the output waveforms of PV system using the proposed OD-PSO MPPT algorithm and the P&O-PSO MPPT algorithm are demonstrated. It takes 0.286 s, 15 iterations, to acquire the GMPP with the P&O-PSO MPPT algorithm. Therefore, the proposed OD-PSO MPPT algorithm has almost the same tracking speed with the firefly MPPT algorithm and has the better MPPT tracking speed comparing to the P&O-PSO MPPT algorithm.

Here, the tracking time is defined as the time from 0 to the moment when the fluctuation of P is less than 5%. In this paper,

TABLE III
SIMULATION RESULTS OF MPPT

	Algorithms	P_{ave}	MPPT efficiency	Iterations	Tracking time
Case I	OD-PSO	134.17(W)	99.82%	8	0.210(s)
	Firefly	129.4(W)	99.75%	8	0.206(s)
	P&O-PSO	134(W)	99.80%	15(PSO)	0.286(s)
Case II	OD-PSO	114.12(W)	99.83%	7	0.185(s)
	Firefly	108.74(W)	99.73%	8	0.206(s)
	P&O-PSO	113.9(W)	99.82%	13(PSO)	0.266(s)

we implemented a threshold value of 5% to indicate that the system has reached its steady state.

In order to thoroughly compare the tracking performances of these three MPPT algorithms, a variable P_{ave} is introduced to represent the average output power of the PV system with the MPPT algorithm during time duration T . P_{ave} can be expressed as

$$P_{ave} = \frac{\int_0^T p dt}{T} = \frac{\sum_{i=1}^{Num} p_i \Delta t}{Num \Delta t}. \quad (9)$$

Term p_i is the instantaneous power, Δt is a constant sampling time, and Num is the total sampling number. Term T , the tracking time, is defined as the duration from the starting of the MPPT algorithm to the moment when the GMPP is captured.

Further, the tracking performances of these three MPPT algorithms can also be described by MPPT efficiency η , which can be calculated as

$$\eta = \frac{P_1}{P_2} \times 100\%. \quad (10)$$

Term P_1 means the output power is in the stable mode of the PV system under the OD-PSO MPPT algorithm. P_2 is the theoretical maximum output power of the PV system under a certain PSC.

Since the simulation data in PSIM can be exported, P_{ave} and the MPPT efficiency can be calculated using (9) and (10), respectively. P_{ave} and the MPPT efficiency with the proposed OD-PSO MPPT algorithm, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm can also be obtained. According to the simulation results, P_{ave} is 134.17, 129.4, and 134 W, while the MPPT efficiency is 99.82%, 99.75%, and 99.80% using the three MPPT methods, respectively, as given in Table III. It is obvious that the proposed OD-PSO MPPT algorithm has the best overall performance considering P_{ave} and the tracking speed.

Case II: The irradiance of PV panels is given in Table I. The P - V characteristic curve is shown in Fig. 4. From Fig. 4, the voltage at GP_2 is 33.85 V and the maximum power is 131.71 W.

The simulation results for Case II are shown in Fig. 13. In Fig. 13(a), the output waveforms of PV system using the proposed OD-PSO MPPT algorithm and the firefly MPPT algorithm are shown. With the proposed OD-PSO MPPT algorithm, it takes 0.185 s, seven iterations, for acquiring the GMPP, wherein five iterations are used for the OD MPPT algorithm and two iterations for the PSO MPPT algorithm. With the firefly

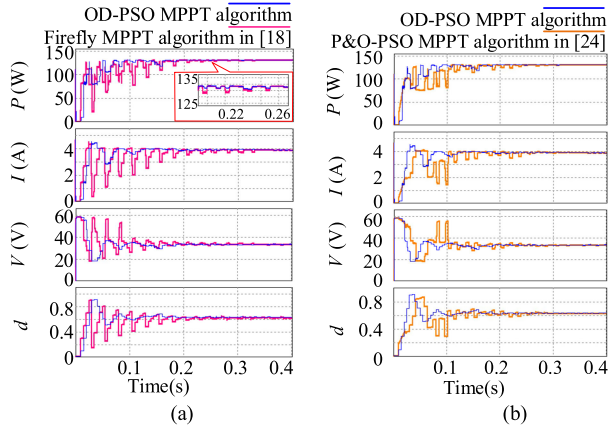


Fig. 13. Output waveforms of the PV system in Case II. (a) OD-PSO MPPT algorithm and the firefly MPPT algorithm in [18]. (b) OD-PSO MPPT algorithm and P&O-PSO MPPT algorithm in [24].

MPPT algorithm, it takes 0.206 s, eight iterations. In Fig. 13(b), the output waveforms of PV system using the proposed OD-PSO MPPT algorithm and the P&O-PSO MPPT algorithm are shown. It takes 0.266 s, 13 iterations, for successfully tracking the GMPP with the P&O-PSO MPPT algorithm.

Additionally, according to the simulation results in PSIM, P_{ave} is 114.12, 108.74, and 113.9 W, while the MPPT efficiency is 99.83%, 99.73%, and 99.82%, using the proposed the OD-PSO MPPT algorithm, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm, respectively, which is given in Table III. It is obvious that the proposed OD-PSO MPPT algorithm also has the best overall performance with the highest P_{ave} and fastest tracking speed in Case II. Furthermore, the power fluctuation with the proposed OD-PSO MPPT algorithm is much lower than that with the firefly MPPT algorithm and the P&O-PSO MPPT algorithm, as shown in Fig. 13. Therefore, the PV system has a better dynamic performance with a higher output power.

Please note that the proposed OD-PSO MPPT method is able to rapidly and accurately track and find the GMPP in various PSCs, regardless of the GMPP position. In this study, without the loss of generality, we have implemented two case studies to test the functionality of the proposed method. The GMPPs in the case studies are designed to locate at the rightmost peak and the middle peak in Case I and Case II, respectively.

The dynamic waveforms of the PV system using the proposed OD-PSO MPPT algorithm are given in Fig. 14. It is obvious that the proposed algorithm can accurately capture the GMPP that varies with different solar irradiances.

V. EXPERIMENTAL RESULTS

In order to verify the feasibility and effectiveness of the OD-PSO MPPT algorithm, an experiment is also conducted. The experiment platform is built, as shown in Fig. 15. The parameters of the circuit in the experiment are the same as those in the simulation, and the sampling time of V and I is 0.075 s. The OD-PSO MPPT algorithm is implemented into a single chip

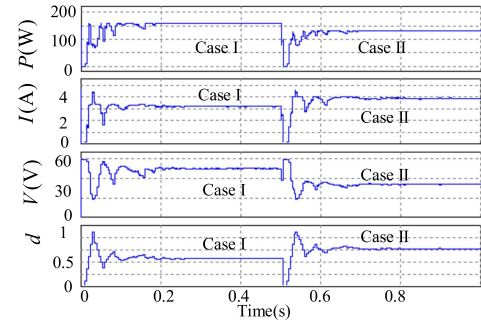


Fig. 14. Dynamic waveforms of the PV system using the OD-PSO MPPT algorithm.

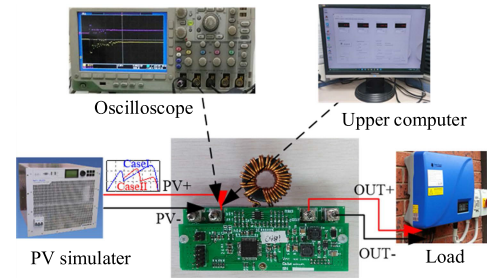


Fig. 15. Experimental platform.

TABLE IV
MAIN EXPERIMENTAL DEVICES

Devices	Types	Parameters
PV simulator	TC.P.32	1000V/13A
Oscilloscope	DPO 4054B	500MHz
Resistive Load	BT ₂ -014	0-10.73Ω
Digital Signal Processor	MC56F8245	60MHz

microprocessor, MC56F8245. In this paper, the PV array is replaced by a PV simulator for higher controllability, which can edit the P - V curve and the I - V curve for Case I and Case II. As mentioned before, the inverter is replaced by a resistive load in this experiment. The main devices in experiment are provided in Table IV. In the experiment, ε is set as 0.025, and γ is set as 0.1 for the OD MPPT algorithm. The parameters of the PSO MPPT algorithm are $\omega = 0.4$, $c_1 = 0.2$, $c_2 = 2$, $j_{max} = 10$, and $v_{max} = 0.035$. The experimental results for Case I and Case II are given in Figs. 16 and 17, respectively.

It is noteworthy that a longer sampling time in the experiment is chosen due to the slow response speed of PV simulator used in our laboratory and the time cost for the buck converter and PV simulator to reach harmonious synergy. The superiority of the proposed OD-PSO MPPT algorithm over established ones in terms of a faster tracking speed and more accurate GMPP location can be noticed in the simulation results. The experiment is for the purpose of verifying the feasibility of the proposed method being implemented into hardware.

Case I: Fig. 16 shows the output power and the current and voltage curves of the PV system in Case I using the proposed OD-PSO MPPT algorithm and using the firefly MPPT algorithm and the P&O-PSO MPPT algorithm, respectively.

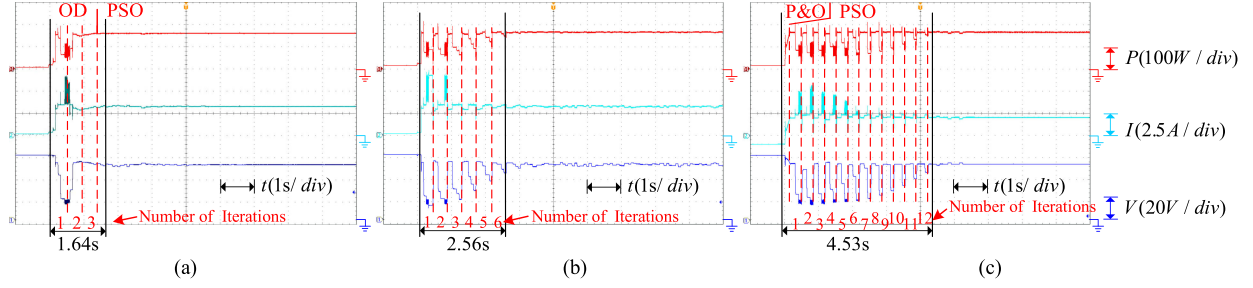


Fig. 16. Output waveforms of the PV system in Case I. (a) OD-PSO MPPT algorithm. (b) Firefly MPPT algorithm. (c) P&O-PSO MPPT algorithm.

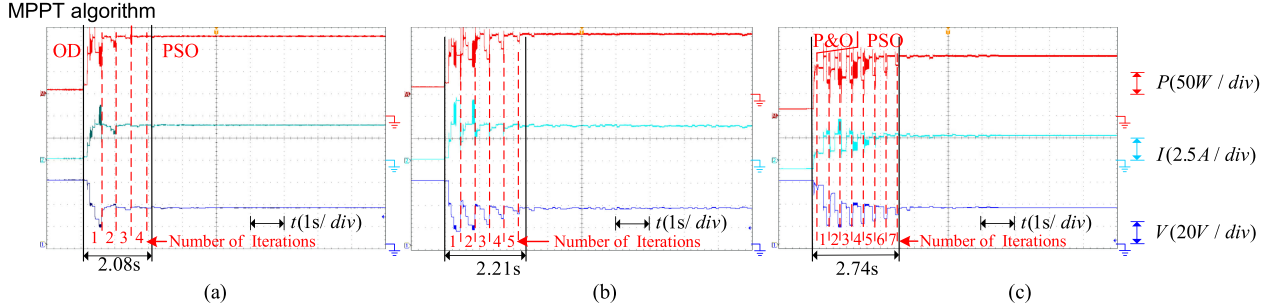


Fig. 17. Output waveforms of the PV system in Case II. (a) OD-PSO MPPT algorithm. (b) Firefly MPPT algorithm. (c) P&O-PSO MPPT algorithm.

To compare the experimental results, P_{ave} , MPPT efficiency, and the tracking time are obtained from the experimental waveforms and the experimental data are exported from an oscilloscope. From Fig. 16, it takes 1.64 s, four iterations to reach the GMPP in Case I using the proposed OD-PSO MPPT algorithm, wherein three iterations using the OD MPPT algorithm. It takes 2.56 s, 6 iterations using the firefly MPPT algorithm and 4.53 s, 12 iterations using the P&O-PSO MPPT algorithm in this case, respectively. It should be noticed that, in addition to the sampling time and the number of particles, the time consumption for each iteration and the tracking time in experiment are actually also influenced by the speed of the employed digital controller and the program statements written for the MPPT algorithms.

Using the same comparison standards as in the simulation, P_{ave} of the PV system with the proposed OD-PSO MPPT algorithm, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm can be obtained by (8). P_{ave} in Case I with the OD-PSO MPPT, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm is 112.58, 126.07, and 133.08 W, respectively.

Case II: Fig. 17 shows the output power and the current and voltage curves of the PV system in Case II using the proposed OD-PSO MPPT algorithm and using the firefly MPPT algorithm and the P&O-PSO MPPT algorithm, respectively.

From Fig. 17, it can be seen that it takes 2.08 s, four iterations to reach the GMPP in Case II using the proposed OD-PSO MPPT algorithm, whereas three iterations using the OD MPPT algorithm and one iteration using the PSO MPPT algorithm. It takes 2.21 s, five iterations using the firefly MPPT algorithm and 2.74 s, seven iterations using the P&O-PSO MPPT algorithm in Case II, respectively.

Average power P_{ave} of the PV system with the proposed OD-PSO MPPT algorithm, the firefly MPPT algorithm, and

TABLE V
EXPERIMENTAL RESULTS OF THREE MPPT ALGORITHMS

	Algorithms	P_{ave}	MPPT efficiency	Iterations	Tracking time
Case I	OD-PSO	133.82(W)	98.29%	3	1.64(s)
	Firefly	126.07(W)	97.79%	6	2.56(s)
	P&O-PSO	133.08(W)	98.18%	12(PSO)	4.53(s)
Case II	OD-PSO	113.46(W)	98.71%	4	2.08(s)
	Firefly	106.58(W)	97.60%	5	2.21(s)
	P&O-PSO	111.63(W)	98.18%	7(PSO)	2.74(s)

the P&O-PSO MPPT algorithm can be calculated by (8). P_{ave} in Case II with the OD-PSO MPPT algorithm, the firefly MPPT algorithm, and the P&O-PSO MPPT algorithm is 110.85, 106.58, and 111.63 W, respectively. Power fluctuation of the OD-PSO MPPT algorithm is noticeably smaller than that with the firefly MPPT algorithm and the P&O-PSO MPPT algorithm, which agrees with simulation results.

The numerical results for the experiment on P_{ave} , the number of iterations, and tracking time are presented in Table V. It can be found that the experimental results are consistent with the simulation results. With the proposed OD-PSO MPPT algorithm, the PV system has a faster GMPP tracking speed, a higher output power, and better dynamic performances.

It should be noticed that according to Figs. 16 and 17, the tracking times are all longer than 2 s in the experiment with all three MPPT methods. This is caused by the fact that the sampling time is set to be 0.075 s in the experiment, which however is 0.005 s in simulation. Normally, the tracking time in Figs. 16 and 17 can satisfy most of the requirements in practical applications where the change rate of solar irradiance is

considered relatively slow [35]. On the other hand, the tracking speed of the OD-PSO MPPT algorithm can be improved significantly by setting a smaller sampling time; however, the cost of digital processors will also be increased for faster calculation speeds and high performances. It is worth noting that in the PSO method, the current levels shift quickly, which may increase the stress on the switches and decrease the control bandwidth to achieve a higher accuracy. In this study, the use of OD method in the proposed OD-PSO MPPT algorithm, to some extent, mitigates the computational stress caused by PSO. This can be observed in Figs. 16 and 17, where the current does not show significant fluctuations. According to Figs. 16(a) and 17(a), the peak values of dv/dt using the OD-PSO MPPT algorithm are 5.85×10^{-2} V/ns in Case I and 5.26×10^{-2} V/ns in Case II, and the peak values of di/dt are 9.43 A/ μ s in Case I and 10.59 A/ μ s in Case II, respectively. Compared to the rating parameters of the MOSFET selected in the experiment [38], the switching stresses of the MOSFET are enough for realizing all the three MPPT algorithms in this paper.

VI. CONCLUSION

In this paper, an OD-PSO MPPT algorithm has been proposed for a PV power system to track the GMPP under PSCs. The OD-PSO MPPT algorithm can rapidly find the small region that contains the GMPP without detailed information about the PV array, and thus has a fast MPPT speed. In addition, the OD-PSO MPPT algorithm can make the PV system output more power and have lower power fluctuation compared to the existing MPPT algorithms. Both simulation and experimental results have validated the effectiveness and superiority of the proposed OD-PSO MPPT algorithm over the established MPPT methods.

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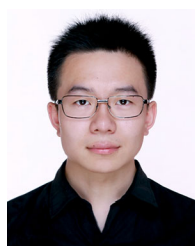
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